

David Ibarra Luna

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SUMMARY

Industrial Chemist with a Master's in Materials Science, specialized in nanomaterials for optoelectronic water sensors applied to industrial water quality assurance. Experienced in interdisciplinary research combining materials development, contaminant analysis, and data processing with Python, aimed at solving real-world environmental challenges.

PROFESSIONAL EXPERIENCE

PhD Student / Research Staff - ICMAB, CSIC (Barcelona, Spain) Sep 2025 – Present

- Research staff in the Nanomol group under the supervision of Marta Mas-Torrent and Núria Crivillers, working on organic materials for electronic and sensing applications.

Research stay at institutes of CSIC (Barcelona, Spain) Sep 2024 – Dec 2024

- Developed carbon nanomaterials for the detection of nitrates in water, contributing to a project with direct applications in optoelectronic sensors.
- Applied advanced material characterization techniques, which included electron microscopy and spectrophotometry, and analyzed and processed spectral data using Python, which expedited the project's interpretation of results.

Robotics teacher (High School) Aug 2023 – Jun 2025

- Taught robotics and programming to high school students using the STEM-based FIRST Lego League methodology, fostering technical and problem-solving skills through hands-on projects.

Laboratory analyst Jun 2020 – Jan 2021

- Performed quantitative analysis of food samples to determine nutritional content, ensuring data accuracy through strict adherence to laboratory protocols.

PROJECTS

Spectrum Data Loader - Python Package (PyPI) [Link to Project](#)

Developed and published a Python library to simplify the loading and initial processing of spectroscopy data files.

EDUCATION

2023–2025 M.Sc. in Materials Chemistry, Autonomous University of Nuevo Leon, (Grade: 94.44/100)

2019–2022 B.Sc. in Industrial Chemistry, Autonomous University of Nuevo Leon, (Grade: 82.69/100)

PUBLICATIONS

Ibarra, David et al. (Apr. 2023). "Synthesis of Graphene Quantum Dots Coupled to Au Nanoparticles: A Facile and Versatile Route Using Different Carbon Sources". In: *C* 9.2, p. 45. DOI: [10.3390/c9020045](#).

- (2026). "A Comparative Study of Synthesis Routes for Nitrogen-Doped Graphene Quantum Dots in Fluorescence Quenching-Based Nitrite Detection". In: *ACS Omega* 11.27, p. 39700. DOI: [10.1021/acsomega.5c12566](#).

AWARDS

2026 ANUIES Master’s Thesis Award – Water Category, for the thesis: *Development of an Optoelectronic Sensor Based on Graphene Quantum Dots for Nitrite Detection in Shallow Well Water.*

SKILLS

Technical Skills	Nanomaterial and Organic Synthesis, Material Characterization, Programming analysis, Quantitative Analysis.
Programming & Software Languages	Python, Git Bash, L ^A T _E X, MS Office Suite, Inkscape, ImageJ. Spanish (Native), English (Advanced), French (Beginner).